Alliant: Getting To Know You

Requirements (First Draft)

Frontend

1. Input for name and keywords that will help guide the search.
    a. Perhaps first iteration give us links to social media profiles
2. Output
    a. Photos of the individuals/maybe a recently posted event
        i. It would be nice to have at least one professional headshot type of photo so they know that it is the correct individual.
    b. AI generated content about the profile
3. Format should be clear and relatively organized. This helps the user experience.
4. Some sort of loading time feature. We could have a rough estimate timeline as well as a more detailed description of what stage the process is at (scraping, data storage, AI generation)
5. Ability to stop the web scraping process if taking too long/the user wants to stop the process. This should kill or undo what the progress was up until that point.
6. Loose web design of page:
    a. Have a search bar and buttons on top
    b. Data generation below that
7. Some sort of Web Application: Could change in the future!

Web Scraping Element

1. We want user pictures to be webscrapped - This could include photos of an individual or photos that are on their own respective social media.
2. We should look at keywords: This is marital/relationship status. This would also be age, general location, and possibly hobbies
3. Keywords should be used for more accurate identity verification. For instance, if two john smiths and one is 20 years older, this data should be used to better verify their identity.
4. If not enough information is given/known about them, we need a way to report back that not enough/any information was found about this individual.
    a. For the first iteration, this could be that we report no Facebook, instagram, or linkedIn profiles were found
5. Start by getting data from Facebook, Instagram, and LinkedIn for example
6. We need to implement some way to cut off if we have enough data about an individual or if that data is too old
    a. For the first iteration, this could be that we don't look at any posts that occur beyond a year ago

Database

1. Storage in a NoSQL database for now (eg. MongoDB).
    a. First iteration suggested categories:
        i. Age
        ii. Name
        iii. Place of Birth
        iv. Past Companies
        v. Keyworded hobbies
        vi. School
        vii. Education majors/minors/certifications
        viii. Photo URLs
        ix. Most recent two or three talking points/posts (This could be used for AI summary generation later on. Important for later iterations to check the dates of certain posts)
2. Table for AI summary output

AI Summary Generation

The following depends a lot on what were using, mostly ChatGPT if budget allows, Llama 3.1 if not:

1. Ability to generate a concise summary given certain data information. This should be in a short paragraph (or two) with no longer than around 500 words.
2. The grammar and flow of the generated content should make sense to the reader.
3. There should be some way of storing the text output for ease of use to the frontend developers. Most likely this could go to the database.

Goals for Agile Development

- Start with scraping to find the data required.
- Want to shoot for some sort of demo/working feature every month or so
- We want to split up development but work on one section at a time. So for now (10/5) this means that we will all be working on web scraping.

Development Notes

10/5: Web Scraping - research selenium web scraping **BUT** APIs for social media sites would be huge. We are going to research both. We also want to look into how chat gpt API works.